

Fortress AI

System & Dashboard Whitepaper

External technical review - Neural Ops UI v2 - Not investment advice

Fortress AI - System & Dashboard Whitepaper

Document purpose: External technical review of the Fortress AI platform, its operational architecture, and every dashboard panel with metric definitions, data provenance, and trading relevance.

Canonical filename: docs/FORTRESS_AI_WHITEPAPER.md

PDF export: docs/FORTRESS_AI_WHITEPAPER.pdf (generate via scripts/generate_whitepaper_pdf.py)

Version: Neural Ops UI v2 (June 2026)

Primary dashboard: Flask app at port 8050 (FORTRESS_AI_DASHBOARD_BUILD / FORTRESS_AI_DASHBOARD_PORT)

Disclaimer: Research and paper-trading infrastructure. Not investment advice.

1. Executive summary

Fortress AI is a multi-agent quantitative trading platform built around four distinct execution paths plus a Classic sibling stack for A/B comparison:

Layer	Agent / service		Role	LLM?
Intraday skim	Skim (fortress-ai-skim-swarm)	Swarm	Always-on rule-based skims across 16 symbols	1-minute long/short No
AI-infra overlay	Infra (fortress-ai-infra-swarm)	Swarm	SRP stack propagation anchored on SMH; L1-L4 adaptive universe	No
Strategic AI	Unified AI (fortress-ai-agent)	Agent	Periodic portfolio decisions on non-skim universe	LLM-driven on Yes (DeepSeek)
Legacy SPY	SPY Intraday Agent		Ladder-based session trading (optional; disabled when skim is primary)	SPY No
Classic sibling	trading-bot repo		Rules-driven multi-agent stack on separate Alpaca account	Optional advisory

The **Command Center dashboard** is the single operational surface for monitoring all paths, diagnosing why trades did or did not occur, tracking P&L, governing self-improvement, and comparing Fortress AI against **Classic** (trading-bot).

Design principles (June 2026):

- Per-symbol independence - each skim/infra ticker owns parameters, pattern disables, and learning state.
- Deterministic hot-path adaptation - live recursive improvement without LLM latency in swarm loops.
- Tiered self-improvement - parameter tuning (Tier 0-1), prompt evolution (Tier 2), immutable risk rails (Tier 3).
- Recursive SI pipeline - integrity scan -> recommendation queue -> agent/human review -> optional autonomous code implementation.
- Full audit trail - every decision, block reason, cost, and governance action is JSONL-logged.

Related docs: docs/INFRA_SWARM.md, docs/UNIFIED_AI.md, docs/SELF_IMPROVEMENT.md, docs/COMPARISON_METRICS.md

2. System architecture

2.1 Data flow

External data (Alpaca, yfinance/IEX, SEC EDGAR, FRED, news sentiment, COT) feeds Skim Swarm, Infra Swarm, Unified AI, and Domain Ingest. Agents write to JSONL/JSON stores under data/. The Flask dashboard (dashboard/ai_command_center.py) aggregates state and serves the Neural Ops UI v2.

Classic Fortress metrics are read-only via utils/classic_bridge.py for comparison and SI capability review.

2.2 Production systemd services

Unit	Role
fortress-ai-dashboard	Neural Ops UI (:8050)
fortress-ai-skim-swarm	1m skim basket
fortress-ai-infra-swarm	AI-infra SRP overlay
fortress-ai-agent	Unified LLM agent
fortress-ai-rth-intraday-si	RTH autonomous SI loop (default 30 min)
fortress-ai-operator-status	15m operator snapshots
fortress-ai-governance	Daily governance maintenance (06:15 ET)

All schedules use America/New_York (FORTRESS_SYSTEM_TZ).

2.3 Refresh tiers

Tier	Interval	Mechanism	Panels affected
SSE	~5s	GET /api/stream/decisions	Header status, AI Mind, positions
Fast	60s + SSE	GET /api/ai/current_state	Market pulse, portfolio, beliefs
Medium	60s	Partial state patch	Domain intel, screener, LLM spend, market consciousness
Skim	45s	GET /api/skim/status	Skim swarm panel
Infra	45s	GET /api/infra/status	Infra swarm panel
Slow	300s	Ingest, SI, governance APIs	Ingest health, SI capability review, prompt evolution

Server-side cache: build_current_state() TTL ~25s.

3. Execution engines

3.1 Skim Swarm - core intraday engine

Universe (default 16 symbols): SPY, NVDA, MSFT, GOOG, AMZN, AAPL, SOXX, NASA, BRK.B, AGIX, AVGO, LLY, V, MA, PLTR, CRWD

Each symbol is an independent agent with:

- **4 patterns:** rip_fade (short), pullback_uptrend (long), momentum_long, momentum_short
- **Per-symbol params:** entry deltas, target multiplier, disable_patterns, side pauses, causation blocks
- **Dual-timescale learning:** 10-year daily historical seeds + live 1-minute session stats
- **Adaptive max open:** utils/adaptive_max_open.py scales effective max_open from market consciousness + session markers (base -> ceiling -> aggressive ceiling)

Recursive self-improvement loop (every 10 exits, then every 5, per symbol):

1. Disable PnL-negative patterns
2. Pause toxic sides (pause_long / pause_short)
3. Full symbol pause if session bleeds (pause_entries)
4. Tighten entry thresholds via causation keys (pattern|side|spy_regime|sym_regime|score_bucket)

3.2 Infra Swarm - AI stack SRP overlay

Purpose: Trade semiconductor / AI supply-chain propagation layers (SRP) anchored on SMH with adaptive L1-L4 universe.

Concept	Detail
Anchor	FORTRESS_INFRA_ANCHOR (default SMH)
Universe	Adaptive rewrite via FORTRESS_INFRA_ADAPTIVE_UNIVERSE
Layers	L1-L4 stack mapping per symbol
Session SI	FORTRESS_INFRA_SWARM_SESSION_SI - pause entries on critical session
Risk	FORTRESS_INFRA_DAILY_STOP_USD, FORTRESS_INFRA_MAX_OPEN_POSITIONS, L1 gross cap

Data: data/infra_swarm/decisions.jsonl, learned/, session_policy.json, adaptive_universe.json

3.3 Unified AI Agent - LLM strategic layer

Orchestrator: agents/unified_ai_agent.py
Execution package (June 2026 refactor):

Module	Role
unified_ai/position_manager.py	Entry dedup, held-qty checks
unified_ai/order_executor.py	Chunked exits under notional cap
unified_ai/risk_controller.py	Oversized position detection
unified_ai/legacy_flattener.py	Periodic trim of legacy oversized lots
unified_ai/settings.py	Config from env + config/default.yaml
utils/unified_enter_guard.py	Enter cooldown + already_holding gate
utils/order_chunking.py	chunk_qtys, held_qty_for_symbol, notional cap

Key env vars:

Variable	Default	Effect
FORTRESS_MAX_ORDER_NOTIONAL_USD	50000	Max single-order notional; triggers chunked exits
POSITION_DEDUPLICATION_ENABLED	true	Block duplicate entries on same symbol
FORTRESS_UNIFIED_ENTER_COOLDOWN_SEC	(guard)	Cooldown between enter attempts

Detectable log markers: already_holding, chunked_exit, enter_cooldown, entry_blocked_by_cooldown, exit_fill_confirmed, exit_unfilled, open_exit_order_pending, insufficient_buying_power_short

Adaptive exit RSI (June 2026): Before each LLM cycle, utils/unified_position_exit.py evaluates open broker positions for deterministic exits - regime-scaled profit targets, stop-loss, RSI overbought take-profit, and EOD flatten. Config: config/unified_position_exit.yaml, env FORTRESS_UNIFIED_POSITION_EXIT. Markers: adaptive_exit_threshold, rsi_overbought_exit, eod_profit_flatten.

Fill-gated ledger: Exits poll Alpaca (utils/alpaca_order_confirm.py) before recording PnL. Unfilled orders return exit_unfilled; ledger rows use exit_fill_confirmed only (never exit_fill_pnl_estimate on submit).

Broker reconciliation: utils/broker_reconciliation.py - integrity codes premature_exit_ledger, operator_broker_open_drift, broker_open_sell_backlog. Hygiene: utils/alpaca_order_hygiene.py cancels phantom/duplicate open SELL orders.

Unified AI does **not** trade skim/infra universe symbols (FORTRESS_AI_SYMBOL_DENYLIST).

3.3a Skim / Infra execution guards (June 2026)

Module	Role
utils/swarm_buying_power.py	Block enter_short when buying power depleted (Alpaca 40310000)
utils/alpaca_execution.py	has_open_exit_order - dedupe exit spam
agents/skim_swarm/act.py, agents/infra_swarm/act.py	Honor open_exit_order_pending before new SELL

Env: FORTRESS_SWARM_SHORT_MIN_BUYING_POWER_USD (default 150), FORTRESS_SWARM_SHORT_BP_GATE

Markers: insufficient_buying_power_short, swarm_short_bp_gate

3.4 Market consciousness

Module: utils/market_consciousness.py

Purpose: Session intent, tape/VIX regime, alpha vs SPY, analogue days, proactive SI triggers. Feeds adaptive max-open and operator diagnostics.

Env: FORTRESS_MARKET_CONSCIOUSNESS, FORTRESS_CONSCIOUSNESS_CACHE_SEC

Data: data/market_consciousness/knowledge.json

3.5 RTH intraday autonomous SI

Script: scripts/rth_intraday_si_loop.sh -> scripts/rth_intraday_si.py -> utils/rth_autonomous_si.py

Each cycle (default 1800s, overridable by capability review):

1. Integrity scan (utils/integrity_diagnostics.py) - includes **broker reconciliation** (premature_exit_ledger, operator_broker_open_drift, broker_open_sell_backlog)
2. SI recommendation queue process
3. Edge scorecard / autofix
4. Swarm session SI
5. Per-symbol tune
6. Governance maintenance hooks

Env: FORTRESS_RTH_INTRADAY_SI, FORTRESS_RTH_SI_INTERVAL_SEC

Artifacts: data/rth_intraday_si/, data/integrity_scan_latest.json

Freeze: Operator halt sets SI-FROZEN: trading_halted - no auto-apply while halted.

3.6 Operator status monitor

Service: fortress-ai-operator-status

Script: scripts/operator_status_report.py

Interval: FORTRESS_OPERATOR_STATUS_INTERVAL_SEC (default 900 = 15 min)

Output: data/operator_status/latest.json, reports.jsonl

Snapshot fields: service health, skim/infra PnL & block mix, adaptive_max_open effective values, SI queue counts, auto-code health, top anomalies, **broker_open_positions** / **broker_symbols** (live Alpaca count - may exceed swarm-reported open when legacy unified-agent holdings exist). Used for independent ops review (no dedicated dashboard panel).

3.7 Recursive SI recommendation queue

Pipeline: integrity scan -> utils/si_recommendation_queue.py -> optional utils/si_code_implementation.py (autonomous code SI)

Disposition	Meaning
auto_resolved	Mitigation already deployed
auto_applied	Tier 0/1 auto-tune applied
monitoring	Watch only
pending_agent_review	Cursor agent triage
pending_human_go	Awaits explicit operator "go"
auto_implement_queued	Autonomous code runner queued

Registry: config/si_fix_registry.json - finding codes, mitigation markers, effort/impact

Data: data/si_recommendation_queue.json, data/si_recommendation_summary.json

APIs:

- GET /api/si/recommendations
- POST /api/si/recommendations/process
- POST /api/si/recommendations/<id>/agent-assess|human-go|implemented
- POST /api/si/code/run, POST /api/si/code/implement/<id>

Autonomous code env: FORTRESS_SI_AUTO_CODE, FORTRESS_SI_AUTO_CODE_MAX_PER_DAY, FORTRESS_SI_AUTO_COMMIT, FORTRESS_SI_AUTO_PUSH, FORTRESS_SI_AUTO_CODE_REQUIRE_E2E, CURSOR_API_KEY

3.8 SI capability review + Classic bridge

Module: utils/si_capability_review.py - cross-stack objectives (Fortress AI + Classic)

Bridge: utils/classic_bridge.py - read-only Classic PnL, fill recency, daily screen health, open positions

Overrides: data/si_capability/overrides.json

APIs: GET /api/si/capability-review, POST /api/si/capability-review/run, GET /api/si/singularity

Classic env: CLASSIC_DATA_DIR, FORTRESS_TRADING_BOT_ROOT, CLASSIC_PNL_LEDGER_PATH

4. Dashboard panels - detailed reference

Template root: dashboard/templates/ai_dashboard.html

JS: dashboard/static/js/ai-dashboard.js

Panel 0: Build stamp banner

Purpose: Deployment verification - confirms the browser loaded the expected UI bundle.

Metric	Source	Why it matters
ui_build	Env FORTRESS_AI_DASHBOARD_BUILD	Prevents reviewing stale UI after deploy

Panel 1: Header bar

Purpose: Operational command center - instance identity, neural state, portfolio headline, operator controls.

Metric	Field	Source	Trading importance
Instance name	state.instance	FORTRESS_INSTANCE_NAME	Multi-VM identification
Mode badge	state.dry_run	FORTRESS_AI_DRY_RUN	DRY-RUN = no live orders
Skim universe count	skim_preview.universe.length	FORTRESS_SKIM_UNIVERSE	Active basket size

Metric	Field	Source	Trading importance
Neural state	state.ui_status	_infer_ui_status()	WAITING ->
			OBSERVING ->
			THINKING ->
			EXECUTING
Portfolio equity	state.portfolio.equity	Alpaca get_account()	Primary capital metric
API spend today	state.today_llm_spend_usd	ai_llm_cost_ledger.jsonl	LLM cost control
Halt button	POST /api/operator/halt	operator_trading_halt.js on	Emergency kill switch
Run AI cycle	POST /api/agent/run-cycle	On-demand agent	unified Manual trigger

Panel 2: Skim Swarm (live)

API: GET /api/skim/status (45s poll)

Template: components/skim_swarm_panel.html

Purpose: Real-time view of the always-on intraday skim basket - primary production path.

Summary tiles

Metric	Source	Trading importance
Daily realized / unrealized / session net	decisions.jsonl + Alpaca	Today's P&L
Exit count	decisions.jsonl	Adaptation sample size
Open positions	Alpaca + state	vs adaptive max_open effective
HALTED	swarm_state.json	Daily stop gate

Per-symbol table

Side, price/change, unrealized/realized, W/L/exits, last action - from learned/*.json and Alpaca.

Panel 3: Infra Swarm (AI stack SRP)

API: GET /api/infra/status (45s poll)

Template: components/infra_swarm_panel.html

Purpose: Monitor AI supply-chain overlay - stack stress, layer assignments, per-symbol P&L.

Metric	Source	Trading importance
Anchor / universe	infra_swarm_config	Active SRP basket
Daily realized	infra decisions + pnl	Infra session P&L
Open positions	latest_metric	vs infra max_open effective
Stack signal / stack_stress	stack_signal	Propagation stress indicator
Per-symbol layer	layer_for_symbol	L1-L4 stack role
dry_run	infra config	Safety mode

Panel 4: Why no trade?

API: trading_diagnostics in current_state; GET /api/trading/diagnostics

Template: components/why_no_trade.html

Purpose: Diagnostic - why blocked/skipped trades occurred (14-day lookback).

Skim column: waves, entry proposed/executed/rate, HALTED, day P&L, **block_reason_counts** (pattern_disabled, pause_entries, spread_too_wide, causation_blocked, cooldown, no_edge,

max_open_positions, eod_force_flatten).

Fortress AI column: dry_run, min_confidence, executed/cycles, block_reason_counts from ai_decisions.jsonl (includes already_holding, entry_blocked_by_cooldown).

Computation: utils/trading_diagnostics.py

Panel 5: Market consciousness

API: GET /api/ai/market_consciousness

Template: components/market_consciousness.html

Purpose: Session-level market read driving adaptive participation and SI triggers.

Metric	Trading importance
session_intent.plan_line	Operator narrative for today's posture
participation_target	How aggressively to trade
consciousness_posture.mode	Risk-on / neutral / defensive
temporal.slot_key	RTH phase (open, mid, close)
self_state.alpha_vs_spy_pct	Fortress vs benchmark
market_tape / vix_regime	Regime context
session_diary entries/exits	Activity vs intent
analogue_days	Historical similarity
market_events	Scheduled catalysts
proactive_si_trigger	Auto-SI escalation flag

Panel 6: AI Mind (column 1)

Template: components/ai_mind.html

Purpose: Unified LLM Agent reasoning, confidence, beliefs.

Market assessment, chain of thought, confidence vs min_confidence gate, trade belief memory (P7), recent decisions - from ai_decisions.jsonl and beliefs/beliefs.json.

Panel 7: Market pulse / Active scan / Positions (column 2)

Template: components/market_view.html

Section	Key metrics	Source
Market pulse	SPY price/change, VIX, RSI(14)	yfinance, charts API
Active scan	Symbol chips, screener source	AI screen_market or Classic fallback
Positions	Sym, Qty, Entry, Last, P&L	Alpaca get_all_positions()

Panel 8: Performance / API usage / System / Domain intel / Ingest (column 3)

Template: components/intelligence.html

LLM spend chart, API cap bar, system checklist (kill switch, Alpaca, SSE), domain intel regime hints, ingest health (SEC/FRED/News/COT).

Panel 9: Self-improvement (Tier 0-1)

APIs: /api/self_improvement/

Template: * components/self_improvement.html

Parameter bounds: confidence_threshold, decision_interval, rsi_entry_threshold. Flow: Propose -> Shadow -> Approve/Reject -> Monitor -> Auto-revert.

Panel 10: SI capability review

APIs: GET /api/si/capability-review, POST /api/si/capability-review/run

Template: components/capability_review.html

Purpose: Cross-stack outcome review vs objectives (Fortress AI + Classic).

Metric	Trading importance
objective_gaps	Unmet SI targets by component
intervention_success_rate	SI effectiveness score
effective_rth_interval_sec	Adaptive RTH SI cadence
stack_metrics	Skim/infra/classic rollup
classic_recommendations	Suggested Classic interventions

Panel 11: Prompt evolution (Tier 2)

APIs: /api/prompt_evolution/

Template: components/prompt_evolution.html

Additive appendix only - cannot modify schema or risk params.

Panel 12: Governance (Tiers 0-3)

APIs: /api/governance/

Template: components/governance_panel.html

Tier	Risk	Behavior
0	Low	Auto-approve if shadow criteria met
1	Medium	24h human veto window
2	Prompt	Human approval via prompt evolution
3	Immutable	Blocked - position size, stops, exposure, pre_trade_gate

Panel 13: Classic vs AI comparison drawer

API: GET /api/comparison

Template: components/comparison_drawer.html

Side-by-side Fortress AI vs Classic on separate Alpaca accounts: equity, P&L, win rate, trades/cycles, opportunity flags, symbol overlap, Classic fill-recency metrics.

Panel 14: Expert mode (overlay)

Trigger: Header Expert or key E | **API:** GET /api/expert/bundle

Tabs: Prompt note, last decision JSON, cost ledger tail, system logs, raw state JSON.

Panel 15: SPY Intraday (not mounted in live UI)

Template and APIs exist; typically disabled when skim swarm is primary.

5. Data architecture

Path	Written by	Read by
data/skim_swarm/decisions.jsonl	Skim agent	Skim panel, diagnostics
data/skim_swarm/edge_scorecard.json	Edge scorecard	RTH SI, dashboard
data/skim_swarm/session_policy.json	Swarm session SI	Entry pause / caps
data/infra_swarm/decisions.jsonl	Infra agent	Infra panel
data/infra_swarm/session_policy.json	Infra session SI	Infra caps
data/ai_decisions.jsonl	Unified AI	AI Mind
data/ai_llm_cost_ledger.jsonl	API costs	API usage
data/beliefs/beliefs.json	Belief manager	Trade belief memory
data/operator_trading_halt.json	Operator	Kill switch
data/operator_status/latest.json	Operator status cron	External ops review
data/si_recommendation_queue.json	SI queue	SI APIs, operator status
data/si_recommendation_summary.json	Queue rollout	Agent triage
data/si_capability/overrides.json	Capability review	Meta-knobs
data/unified_ai/enter_guard.json	Enter guard	Cooldown state
data/market_consciousness/knowledge.json	Market consciousness	Dashboard panel
data/integrity_scan_latest.json	Integrity scan	RTH SI
data/pnl_ledger.jsonl	Unified AI / backfill	Broker reconciliation
config/unified_position_exit.yaml	Adaptive exit RSI thresholds	Unified agent
config/si_fix_registry.json	Fix registry	Deployment detection

6. Security and risk controls

Control	Mechanism
Kill switch	operator_trading_halt.json
Daily stop	FORTRESS_SKIM_DAILY_STOP_USD / FORTRESS_INFRA_DAILY_STOP_USD
Session SI critical	pause_new_entries - no new swarm entries; exits continue
Adaptive max open	Scales skim/infra caps from consciousness + session markers
Orphan universe guard	Entries blocked outside FORTRESS_*_UNIVERSE
Edge gates	RR / cost / expectancy (FORTRESS_EDGE_*)
Position deduplication	already_holding + enter cooldown (Unified AI)
Chunked exits	FORTRESS_MAX_ORDER_NOTIONAL_USD splits oversized orders
Exit fill confirmation	PnL ledger only after Alpaca fill poll (exit_fill_confirmed)
Exit order dedup	open_exit_order_pending blocks duplicate SELL spam
Short BP gate	Skim/infra block shorts when buying power below floor
Legacy flatten	RiskController every 5 min on Unified AI boot path
Bracket tick clamp	Alpaca min \$0.01 offset on bracket stops
Spread filter	FORTRESS_SKIM_MAX_SPREAD_BPS (25)
EOD flatten	Force flat before close
Weekly LLM cap	FORTRESS_AI_WEEKLY_COST_CAP_USD

Control	Mechanism
Symbol denylist	Unified blocked from skim/infra universe
Confidence floor lock	FORTRESS_AI_CONFIDENCE_FLOOR_LOCK
SI freeze on halt	RTH SI skips auto-apply when operator halt active
Infra L1 gross cap	Stack-level long exposure limit

Protected (immutable): `utils/pre_trade_gate.py`, `utils/operator_halt.py` - SI cannot weaken.

7. Known limitations

1. Classic and Fortress AI use **separate Alpaca accounts** with independent capital - skim `max_open_positions` is not constrained by Classic holdings.
2. No billing module - LLM cost tracking only.
3. SPY panel unwired when skim is primary.
4. Historical verify uses daily bars; live skim uses 1-minute bars.
5. Paper trading on Alpaca paper accounts.
6. Infra `/api/infra/status` may return partial data when service is idle.
7. **E2E tests** mock broker fills - production drift (ACCEPTED-but-unfilled, screener vs entry RSI mismatch) is caught by **integrity reconciliation scans**, not unit tests alone.

8. External review checklist

Reviewers should evaluate:

- **Architecture:** Is four-path separation (skim / infra / unified / classic) clean? Are agent responsibilities modular?
- **Risk:** Are dedup, chunked exits, fill-gated ledger, and adaptive max-open coherent on the Fortress AI account (separate from Classic)?
- **Broker truth:** Does operator status `broker_open_positions` align with dashboard Alpaca feed? Any premature `exit_ledger` findings?
- **SI governance:** Is the queue workflow (agent review -> human go -> implement) sufficient before code changes ship?
- **Execution:** Are `block_reason` markers sufficient for postmortems?
- **Reliability:** Do operator status + RTH SI cycles provide enough visibility without dashboard overload?
- **Comparison:** Is Classic bridge data sufficient for fair A/B review?
- **Verification:** Run `./scripts/e2e_verify.sh` before deploy; confirm `data/integrity_scan_latest.json` has no high-severity broker reconciliation findings.

9. Glossary

Term	Definition
Pattern	<code>rip_fade</code> , <code>pullback_uptrend</code> , <code>momentum_long</code> , <code>momentum_short</code>
Winning pattern	Positive net PnL after min samples - not trade win rate
SRP / Infra	Semiconductor supply-chain propagation overlay
Causation key	Context bucket blocking repeated losing entries
Belief (P7)	Persistent learned fact from closed trades
Neural state	WAITING / OBSERVING / THINKING / EXECUTING
SI queue	Recursive self-improvement recommendation workflow
<code>chunked_exit</code>	Exit split into child orders under notional cap
<code>exit_fill_confirmed</code>	Ledger row recorded only after Alpaca reports fill <code>qty > 0</code>

Term	Definition
adaptive_rsi	Unified exit monitor uses regime + RSI for take-profit thresholds
broker_reconciliation	Integrity scan comparing Alpaca, ledger, and operator counts

Fortress AI - Research dashboard - Not investment advice